

## GREAT DUN FELL NO2, United Kingdom

WMO: 032270

Lat: 54.6842N

Lon: 2.4514W

Elev: 2779

StdP: 13.28

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 19.1       | 20.9 | 6.8                           | 8.6 | 37.5 | 15.2 | 13.2 | 27.1 | 103.6                 | 32.3 | 91.4 | 31.7 | 28.7                  | 340  | 1.296 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.0                    | 64.0            | 56.3 | 60.9 | 55.0 | 58.0 | 53.7 | 58.6                | 61.2 | 56.9 | 59.3 | 55.1 | 57.2 | 15.8                 | 190  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 57.8                            | 79.1 | 59.1 | 55.9 | 73.7 | 57.3 | 54.2 | 69.4 | 55.7 | 26.8          | 61.6 | 25.5 | 59.5 | 24.3 | 57.3 | 82.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |     |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-----|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |     |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p) |      |     |
| (4) 84.2          | 70.3 | 58.1 | DB                         | 16.9 | 71.1               | 3.7 | 5.3                                                | 14.2 | 74.9       | 12.1 | 78.0       | 10.0 | 81.0       | 7.3 | 84.9 | (4) |
| (5)               |      |      | WB                         | 16.2 | 64.4               | 2.6 | 6.6                                                | 14.3 | 69.1       | 12.7 | 73.0       | 11.2 | 76.6       | 9.3 | 81.4 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 39.9   | 31.5  | 31.8  | 33.6  | 37.7  | 42.3  | 46.5  | 50.1  | 49.7  | 46.3  | 41.2  | 36.0  | 32.1 |
|                                                                                  | DBStd       | 8.62   | 5.10  | 5.51  | 5.65  | 6.19  | 6.20  | 5.10  | 5.11  | 4.54  | 4.44  | 5.02  | 5.37  | 5.61 |
|                                                                                  | HDD50       | 3845   | 574   | 508   | 510   | 372   | 254   | 132   | 60    | 58    | 128   | 274   | 420   | 555  |
|                                                                                  | HDD65       | 9146   | 1039  | 928   | 974   | 819   | 704   | 556   | 462   | 475   | 562   | 737   | 870   | 1020 |
|                                                                                  | CDD50       | 175    | 0     | 0     | 0     | 3     | 15    | 26    | 64    | 49    | 16    | 2     | 0     | 0    |
|                                                                                  | CDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH74       | 2      | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 27.3   | 33.3  | 33.0  | 30.1  | 24.6  | 23.1  | 21.6  | 21.1  | 23.5  | 23.8  | 30.2  | 31.7  | 31.8 |
| Precipitation                                                                    | PrecAvg     | 53.8   | 6.3   | 4.8   | 3.8   | 3.2   | 3.2   | 3.1   | 3.5   | 4.0   | 4.1   | 5.0   | 5.8   | 6.8  |
|                                                                                  | PrecMax     | 66.9   | 10.0  | 11.7  | 7.2   | 5.9   | 5.9   | 7.0   | 7.9   | 9.6   | 7.5   | 9.8   | 14.5  | 17.7 |
|                                                                                  | PrecMin     | 39.3   | 1.3   | 0.9   | 1.1   | 1.5   | 0.9   | 1.0   | 1.9   | 0.9   | 1.0   | 1.9   | 2.7   | 2.1  |
|                                                                                  | PrecStd     | 7.0    | 2.4   | 3.0   | 1.3   | 1.2   | 1.4   | 1.5   | 1.3   | 1.8   | 1.5   | 2.0   | 2.6   | 3.3  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 47.4  | 49.2  | 52.3  | 57.8  | 63.8  | 65.4  | 68.4  | 67.6  | 61.5  | 55.6  | 52.2  | 48.9 |
|                                                                                  |             | MCWB   | 37.2  | 41.2  | 42.7  | 47.4  | 55.1  | 55.2  | 59.3  | 57.7  | 57.4  | 50.5  | 45.6  | 39.8 |
|                                                                                  | 2%          | DB     | 43.2  | 44.2  | 46.5  | 52.9  | 58.7  | 60.7  | 63.9  | 63.2  | 57.3  | 52.7  | 48.4  | 44.5 |
|                                                                                  |             | MCWB   | 42.2  | 40.9  | 42.0  | 47.2  | 51.5  | 53.6  | 56.2  | 56.8  | 54.4  | 50.6  | 47.1  | 43.8 |
|                                                                                  | 5%          | DB     | 40.7  | 41.5  | 43.7  | 49.5  | 55.1  | 56.9  | 60.9  | 58.8  | 54.7  | 50.5  | 45.9  | 42.1 |
|                                                                                  |             | MCWB   | 40.3  | 40.1  | 41.4  | 45.2  | 49.5  | 52.5  | 55.4  | 55.1  | 52.3  | 49.0  | 45.1  | 41.5 |
|                                                                                  | 10%         | DB     | 38.9  | 39.4  | 41.2  | 46.2  | 51.2  | 53.9  | 57.8  | 55.9  | 52.6  | 48.6  | 43.5  | 40.2 |
|                                                                                  |             | MCWB   | 38.6  | 38.4  | 39.9  | 43.9  | 47.2  | 50.9  | 54.2  | 53.7  | 50.7  | 47.5  | 43.0  | 39.6 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 44.0  | 45.9  | 45.9  | 51.6  | 57.6  | 59.8  | 62.5  | 60.2  | 58.8  | 53.9  | 49.6  | 46.5 |
|                                                                                  |             | MCDB   | 44.2  | 46.6  | 46.8  | 53.6  | 61.3  | 62.2  | 64.8  | 63.4  | 60.5  | 54.2  | 49.6  | 46.9 |
|                                                                                  | 2%          | WB     | 41.9  | 42.5  | 43.8  | 48.6  | 53.8  | 55.4  | 58.8  | 57.9  | 55.8  | 51.3  | 47.2  | 43.8 |
|                                                                                  |             | MCDB   | 42.1  | 43.0  | 45.3  | 50.8  | 57.9  | 58.2  | 61.1  | 61.0  | 56.6  | 51.6  | 47.4  | 44.0 |
|                                                                                  | 5%          | WB     | 39.8  | 40.4  | 41.8  | 46.3  | 50.4  | 53.3  | 57.0  | 56.1  | 53.3  | 49.3  | 45.1  | 41.6 |
|                                                                                  |             | MCDB   | 40.0  | 41.0  | 43.1  | 48.5  | 53.3  | 55.5  | 59.4  | 58.5  | 54.1  | 49.7  | 45.2  | 41.7 |
|                                                                                  | 10%         | WB     | 37.6  | 38.6  | 39.9  | 44.3  | 48.2  | 51.3  | 54.9  | 54.5  | 51.3  | 47.5  | 43.0  | 39.7 |
|                                                                                  |             | MCDB   | 37.8  | 39.2  | 40.8  | 46.4  | 50.6  | 53.5  | 57.1  | 56.1  | 52.1  | 48.0  | 43.1  | 39.9 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 6.6   | 6.6   | 7.0   | 8.1   | 8.8   | 8.1   | 8.0   | 7.2   | 7.0   | 6.5   | 6.5   | 6.5  |
|                                                                                  |             | MCDBR  | 9.7   | 9.3   | 9.7   | 12.4  | 12.5  | 11.7  | 11.8  | 11.1  | 8.9   | 9.4   | 9.6   | 9.3  |
|                                                                                  |             | MCWBR  | 9.1   | 7.2   | 8.1   | 9.0   | 10.0  | 9.6   | 9.4   | 8.5   | 7.6   | 9.5   | 9.5   | 8.9  |
|                                                                                  | 5% WB       | MCDBR  | 9.2   | 7.8   | 8.6   | 9.4   | 11.4  | 10.6  | 10.5  | 9.4   | 8.2   | 8.8   | 9.3   | 9.0  |
|                                                                                  |             | MCWBR  | 9.0   | 7.7   | 8.2   | 8.2   | 9.9   | 9.5   | 9.3   | 8.0   | 8.1   | 9.2   | 9.4   | 9.0  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.286  | 0.299 | 0.323 | 0.341 | 0.343 | 0.342 | 0.343 | 0.333 | 0.318 | 0.306 | 0.284 | 0.269 |      |
|                                                                                  | taud        | 2.337  | 2.436 | 2.390 | 2.356 | 2.383 | 2.399 | 2.395 | 2.436 | 2.487 | 2.496 | 2.408 | 2.295 |      |
|                                                                                  | Ebn at Noon | 212    | 254   | 272   | 281   | 286   | 287   | 284   | 281   | 271   | 248   | 213   | 190   |      |
|                                                                                  | Edh at Noon | 17     | 22    | 29    | 34    | 35    | 35    | 35    | 31    | 26    | 20    | 16    | 14    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 175    | 386   | 736   | 1177  | 1530  | 1535  | 1483  | 1196  | 849   | 456   | 224   | 134   |      |
|                                                                                  | RadStd      | 23     | 43    | 71    | 118   | 116   | 150   | 155   | 89    | 74    | 55    | 23    | 13    |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (57 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page